Exercises learning task 16

2026 – 2027

Exercise 9.42: A melting point test of $n = 15$ samples of a binder used in manufacturing a rocket propellant resulted in $\bar{x} = 68\,^{\circ}\mathrm{C}$. Assume that the melting point is normally distributed with $\sigma = 0.83\,^{\circ}\mathrm{C}$.

(a) Test $H_0 : \mu = 68.3$ versus $H_1 : \mu \neq 68.3$ using $\alpha = 0.01$.

(b) What is the p-value for this test?

(c) What is the β-error if the true mean is $\mu = 65.5$?

Exercise 9.47: Medical researchers have developed a new artificial heart constructed primarily of titanium and plastic. The heart will last and operate almost indefinitely once it is implanted in the patient's body, but the battery pack needs to be recharged about every four hours. A random sample of 50 battery packs is selected and subjected to a life test. The average life of these batteries is $4.05\,\mathrm{h}$. Assume that battery life is normally distributed with standard deviation $\sigma = 0.2\,\mathrm{h}$.

(a) Is there evidence to support the claim that mean battery life exceeds $4\,\mathrm{h}$? Use $\alpha = 0.05$.

(b) What is the p-value for the test in part (a)?

(c) Compute the power of the test if the true mean battery life is $4.5\,\mathrm{h}$.

Exercise 9.51: For the hypothesis test $H_0 : \mu = 7$ against $H_1 : \mu \neq 7$ with variance unknown and $n = 25$, approximate the p-value for each of the following test statistics.

(a) $t_0 = 2.05$ (b) $t_0 = -1.84$ (c) $t_0 = 0.4$

Exercise 9.59: A 1992 article in the *Journal of the American Medical Association* ("A Critical Appraisal of 98.6 Degrees F, the Upper Limit of the Normal Body Temperature, and Other Legacies of Carl Reinhold August Wunderlich") reported body temperature, gender, and heart rate for a number of subjects. The body temperatures for 25 female subjects follow: 97.8, 97.2, 97.4, 97.6, 97.8, 97.9, 98.0, 98.0, 98.0, 98.1, 98.2, 98.3, 98.3, 98.4, 98.4, 98.4, 98.5, 98.6, 98.6, 98.7, 98.8, 98.8, 98.9, 98.9, and 99.0.

(a) Test the hypothesis $H_0 : \mu = 98.6$ versus $H_1 : \mu \neq 98.6$, using $\alpha = 0.05$. Find the p-value.

(e) Explain how the question in part (a) could be answered by constructing a two-sided confidence interval on the mean female body temperature.

Exercise 9.60: Cloud seeding has been studied for many decades as a weather modification procedure (for an interesting study of this subject, see the article in *Technometrics*, "A Bayesian Analysis of a Multiplicative Treatment Effect in Weather Modification," Vol. 17, pp. 161–166). The rainfall in cubic-meters from 20 clouds that were selected at random and seeded with silver nitrate follows: 22.2×10^3, 37.9×10^3, 24.4×10^3, 33.4×10^3, 27.5×10^3, 23.2×10^3, 39.2×10^3, 28.9×10^3, 26.1×10^3, 34.4×10^3, 39.3×10^3, 33.4×10^3, 30.8×10^3, 30.4×10^3, 33.2×10^3, 26.9×10^3, 36.2×10^3, 42.9×10^3, 32.9×10^3, and 38.9×10^3.

(a) Can you support a claim that the mean rainfall from seeded clouds exceeds $30 \times 10^3\,\mathrm{m}^3$? Use $\alpha = 0.01$. Find the p-value.

(e) Explain how the question in part (a) could be answered by constructing a one-sided confidence bound on the mean rainfall in cubic metres.

Exercise 9.126: Consider the following computer output.

One-Sample Z:
Test of $\mu = 26$ vs $\mu > 26$
The assumed standard deviation $= 1.7$

Variable	n	Mean	StDev	SE Mean	Z	p
X	?	26.541	2.032	0.401	?	?

(a) Fill in the missing information.

(b) Is this a one-sided or a two-sided test?

(c) What are your conclusions if $\alpha = 0.05$?

(d) Find a $95\,\%$ two-sided CI on the mean.

Exercise 9.127: Consider the following computer output.

One-Sample T:
Test of $\mu = 100$ vs $\mu \neq 100$

Variable	n	Mean	StDev	SE Mean	$95\,\%$ CI	t	p
X	16	98.33	4.61	?	(?,?)	?	?

(a) How many degrees of freedom are there on the t-statistic?

(b) Fill in the missing information. You may use bounds on the p-value.

(c) What are your conclusions if $\alpha = 0.01$?

(d) What are your conclusions if the hypothesis is $H_0 : \mu = 100$ versus $H_0 : \mu > 100$?